# Mercury-202

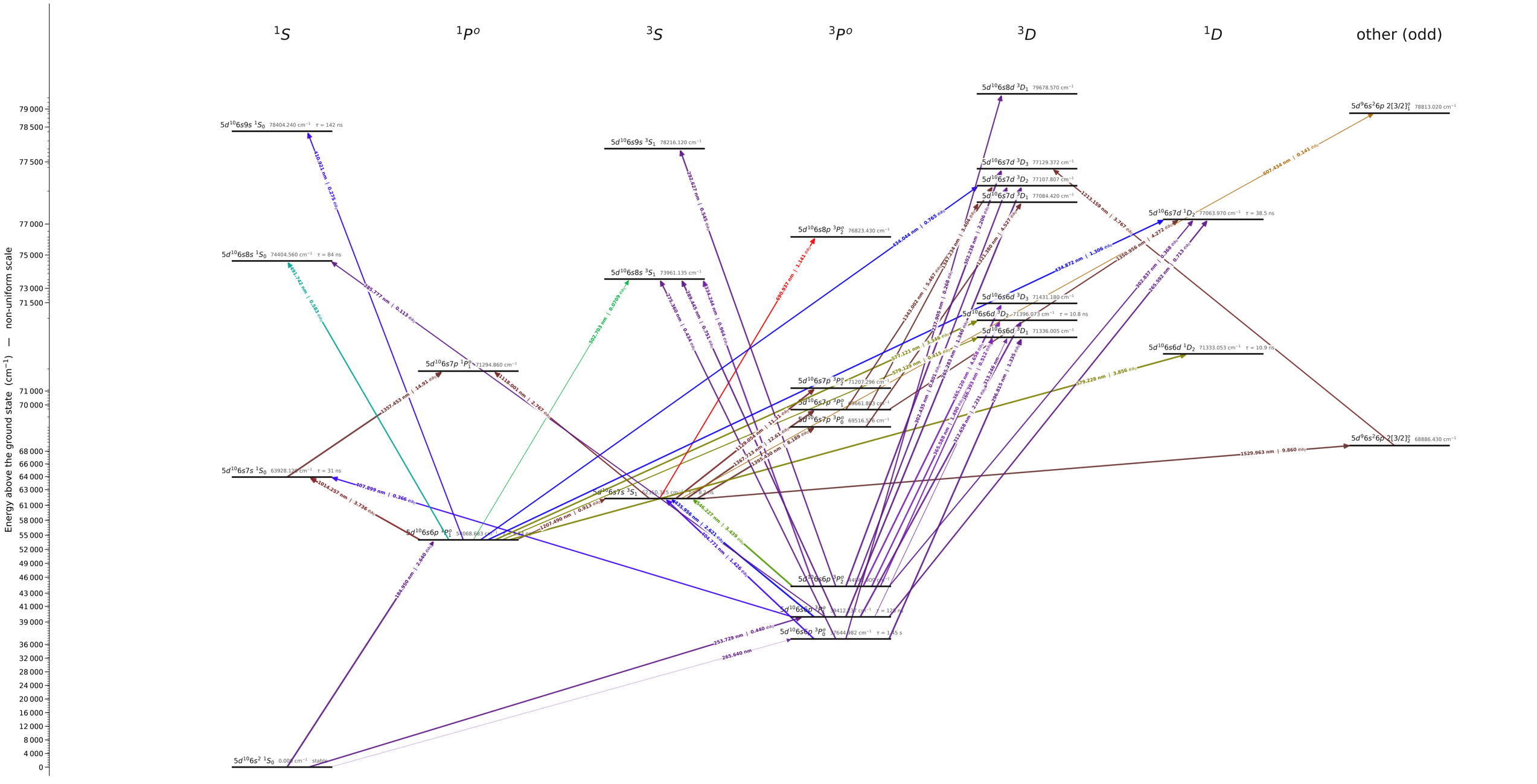
$^{202}$Hg (Hg I), nuclear spin $I = 0$. Strongest classified transitions below 2 µm (NIST): 27 levels, 47 lines.

Hg$^{+}$ ionisation limit 84184.15 cm$^{-1}$ = 10.43750 eV (118.787 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in $ea_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient $A$. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000




## Key transitions

| Transition | λ vac (nm) | λ air (nm) | ν (THz) | $\|\langle J\|er\|J'\rangle\|$ ($ea_0$) | A (s$^{-1}$) | Γ/2π up. (MHz) | branch (%) | Use |
|---|---|---|---|---|---|---|---|---|
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^1P_1^o$ | 184.9499 | – | 1620.9383 | 2.640 | 7.440e+08 | 118.8 | 99.7 | 184.9 nm resonance line |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.7291 | 253.6528 | 1181.545677 | 0.440 | 7.992e+06 | 1.272 | 100 | 253.7 nm intercombination line: MOT / cooling, optical pu… |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_0^o$ | 265.6397 | 265.5606 | 1128.5682 | – | 6.920e-01 | 1.101e-07 | 100 | 265.6 nm optical lattice clock transition |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s8s\ {}^1S_0$ | 285.7770 | 285.6930 | 1049.0434 | 0.113 | 1.100e+06 | 1.895 | 9.24 | 285.7 nm |
| $5d^{10}6s6p\ {}^3P_0^o - 5d^{10}6s6d\ {}^3D_1$ | 296.8150 | 296.7283 | 1010.0315 | 1.335 | 4.600e+07 | – | – | 296.7 nm |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_2$ | 312.6579 | 312.5673 | 958.8513 | 2.231 | 6.600e+07 | 14.74 | 71.3 | 312.7 nm |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_1$ | 313.2462 | 313.1555 | 957.0505 | – | – | – | – | 313.2 nm, optical pumping out of the 253.7 nm MOT |
| $5d^{10}6s6p\ {}^3P_2^o - 5d^{10}6s6d\ {}^3D_3$ | 365.1198 | 365.0158 | 821.0797 | 4.658 | 1.290e+08 | – | – | 365.0 nm |
| $5d^{10}6s6p\ {}^3P_0^o - 5d^{10}6s7s\ {}^3S_1$ | 404.7707 | 404.6564 | 740.6476 | 1.426 | 2.070e+07 | 19.41 | 17 | 404.7 nm |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s7s\ {}^1S_0$ | 407.8988 | 407.7837 | 734.9677 | 0.366 | 4.000e+06 | 5.134 | 12.4 | 407.8 nm |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s7s\ {}^3S_1$ | 435.9561 | 435.8336 | 687.6666 | 2.621 | 5.600e+07 | 19.41 | 45.9 | 435.8 nm |
| $5d^{10}6s6p\ {}^1P_1^o - 5d^{10}6s8s\ {}^1S_0$ | 491.7418 | 491.6045 | 609.6543 | 0.583 | 5.800e+06 | 1.895 | 48.7 | 491.6 nm |
| $5d^{10}6s6p\ {}^3P_2^o - 5d^{10}6s7s\ {}^3S_1$ | 546.2267 | 546.0750 | 548.8425 | 3.439 | 4.900e+07 | 19.41 | 40.2 | 546.1 nm |
| $5d^{10}6s6p\ {}^1P_1^o - 5d^{10}6s7s\ {}^1S_0$ | 1014.2567 | 1013.9787 | 295.5785 | 3.736 | 2.710e+07 | 5.134 | 84 | 1014 nm |
| $5d^{10}6s6p\ {}^1P_1^o - 5d^{10}6s7s\ {}^3S_1$ | 1207.4900 | 1207.1596 | 248.2774 | 0.913 | 3.200e+05 | 19.41 | 0.26 | 1207 nm intercombination branch of 7s 3S1 |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition | λ vac (nm) | Isotopes | Shift (MHz) |
|---|---|---|---|
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 196-198 | 4179.95(3) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 200-198 | -4805.52(4) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 202-198 | -10101.224(22) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 204-198 | -15311.60(4) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 199-198 | -652(6) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_1^o$ | 253.729 | 201-198 | -6410(6) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_0^o$ | 265.640 | 199-201 | 5729.668(8) |
| $5d^{10}6s^2\ {}^1S_0 - 5d^{10}6s6p\ {}^3P_0^o$ | 265.640 | 198-199 | 654.48 |
| $5d^{10}6s6p\ {}^3P_2^o - 5d^{10}6s6d\ {}^3D_2$ | 312.658 | 196-198 | -289.4(12) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_2$ | 312.658 | 200-198 | 328.0(6) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_2$ | 312.658 | 202-198 | 686.7(6) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_2$ | 312.658 | 204-198 | 1038.3(6) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_1$ | 313.246 | 196-198 | -307.5(10) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_1$ | 313.246 | 200-198 | 347.2(6) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_1$ | 313.246 | 202-198 | 725.5(6) |
| $5d^{10}6s6p\ {}^3P_1^o - 5d^{10}6s6d\ {}^3D_1$ | 313.246 | 204-198 | 1097.8(6) |

Shift = ν(first isotope) − ν(second isotope).